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$$\sum_{n=1}^{\infty} \frac{1}{n \cdot \ln \left(1 + \frac{1}{n}\right)} = \sum_{n=1}^{\infty} \frac{1}{\ln \left(1 + \frac{1}{n}\right)^n}$$

negatief criterium:

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{1}{\ln \left(1 + \frac{1}{n}\right)^n} \\ &= \frac{1}{\lim_{n \rightarrow \infty} \ln \left(1 + \frac{1}{n}\right)^n} \\ &= \frac{1}{\ln \left(\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n\right)} \\ &= \frac{1}{\ln(e)} = 1 \neq 0 \\ &\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n \cdot \ln \left(1 + \frac{1}{n}\right)} \text{ is divergent} \end{aligned}$$

References